Chapter 7 — Reaction Kinetics

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Collision theory, activation energy and rate of reaction

Q.1 Explain collision theory, activation energy and the measurement of rate of reaction.

Reaction kinetics is the study of reaction rates. According to collision theory, reactant particles must collide with sufficient energy and proper orientation to react. The minimum energy needed for an effective collision is the activation energy, E_a. Most collisions are ineffective, which is why many reactions are slow.

Rate of reaction is the change in concentration of a reactant or product per unit time. For A → B, rate = $-\Delta[A]/\Delta t = +\Delta[B]/\Delta t$. Units are usually mol dm^{-3} s^{-1}. The average rate is measured over a time interval, while the instantaneous rate is found from the slope of a tangent to a concentration-time curve.

Textbook check (pp. 142–144): Sample Problem 7.1 prints $N_2(aq)$ in the Haber equation; nitrogen should be $N_2(g)$. On p. 144 the calculated rate is 4×10^{-4} mol dm^{-3} s^{-1}, but the following sentence incorrectly says 2.5 mol dm^{-3} per second.

Important Conceptual Short Questions — Topic 1

1. Why do not all collisions lead to reaction?

Only collisions with energy equal to or greater than activation energy and with proper orientation can break old bonds and form new ones. Other collisions simply bounce apart.

2. How does activation energy affect the rate of a reaction?

A high activation energy means fewer particles have enough energy to react, so the rate is low. A lower activation energy allows more effective collisions and a faster rate.

3. Why does the instantaneous rate usually decrease as a reaction proceeds?

Reactant concentration falls with time, so there are fewer effective collisions per second. The concentration-time curve becomes less steep.

4. How can the rate of a reaction be measured experimentally?

Measure a property that changes with concentration, such as gas volume, pressure, colour, pH, conductivity or mass, at regular time intervals.

Topic 2 — Factors affecting rate and Boltzmann distribution

Q.2 Explain the effects of concentration, temperature, surface area and catalyst on reaction rate.

Increasing concentration increases the number of particles per unit volume, so effective collisions become more frequent. For gases, increasing pressure has a similar effect. Increasing surface area of a solid exposes more particles to collision.

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Increasing temperature raises the average kinetic energy and, more importantly, greatly increases the fraction of molecules with energy above E_a. The Boltzmann distribution shifts to higher energy and flattens. A catalyst provides an alternative pathway with lower activation energy and is not consumed overall.

Important Conceptual Short Questions — Topic 2

1. How does temperature affect the Boltzmann distribution?

At higher temperature the curve becomes broader and flatter, and the area under the curve beyond the activation energy increases. More molecules can react effectively.

2. Why does powdered calcium carbonate react faster than marble chips?

Powder has a much larger surface area, so more acid particles can collide with the solid at the same time.

3. How does a catalyst increase reaction rate without being consumed?

It provides an alternative reaction pathway with lower activation energy. It may take part in intermediate steps but is regenerated at the end.

4. Why does increasing pressure increase the rate of a gaseous reaction?

Higher pressure compresses the gas, increasing concentration. More frequent collisions lead to a higher rate.

Topic 3 — Rate law, order and molecularity

Q.3 Explain rate law, rate constant, order of reaction and molecularity.

For a reaction, rate = $k[A]^m[B]^n$, where k is the rate constant and m and n are orders determined experimentally. The overall order is m + n. Order can be zero, integer or fractional. Molecularity is the number of reacting species in an elementary step; it is always a whole number and applies only to a single step.

For a zero-order reaction, rate is independent of concentration. For a first-order reaction, rate is proportional to concentration. For a second-order reaction, rate is proportional to the square of concentration or the product of two concentrations. The units of k depend on overall order.

Order	Rate law	Effect of doubling [A]	Units of k
Zero	rate = k	No change	mol dm^{-3} s^{-1}
First	rate = k[A]	Rate doubles	s^{-1}
Second	rate = $k[A]^2$	Rate quadruples	dm^3 mol^{-1} s^{-1}

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Important Conceptual Short Questions — Topic 3

1. Why can order be zero or fractional but molecularity cannot?

Order is an experimental quantity obtained from the rate law, so it may have any value. Molecularity counts particles in one elementary step, so it must be a whole number.

2. How does a rate law show the order of a reaction?

The power of each concentration term gives the order with respect to that reactant. The sum of the powers gives the overall order.

3. Why is the rate constant independent of concentration but dependent on temperature?

k is a proportionality constant for a given reaction at a given temperature. Changing concentration changes rate, not k; changing temperature changes the fraction of effective collisions and therefore k.

4. How can the units of k indicate the overall order?

Since rate has units mol dm^{-3} s^{-1}, the units of k must balance the concentration powers in the rate law. Different overall orders therefore give different units.

Topic 4 — Determination of rate constant and reaction mechanism

Q.4 Describe how the rate constant is determined and how a reaction mechanism is related to the rate law.

The initial-rate method compares initial rates while changing one reactant concentration at a time. The order with respect to each reactant is found, then k is calculated from rate = $k[A]^m[B]^n$. For a first-order reaction, the half-life method uses $k = 0.693/t_{1/2}$.

A reaction mechanism is the sequence of elementary steps. The slowest step is the rate-determining step. The rate law predicted by the mechanism must match the experimental rate law. Species taking part only after the slow step do not appear in the rate law.

Important Conceptual Short Questions — Topic 4

1. Why is the slowest step called the rate-determining step?

The overall reaction cannot proceed faster than its slowest elementary step, just as a narrow road limits traffic flow.

2. How can initial-rate data be used to find the order with respect to a reactant?

Keep other concentrations constant and change one concentration. If doubling it doubles the rate, the order is 1; if it quadruples the rate, the order is 2; if there is no change, the order is 0.

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3. Why does a first-order reaction have a constant half-life?

For first-order decay, the time needed for the concentration to fall to half is independent of the starting concentration because rate is proportional to the amount present.

4. How can a proposed mechanism be tested against a rate law?

Write the rate law from the slow step and any fast equilibrium steps before it. If it matches the experimental rate law, the mechanism is consistent with the data.

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